

Yu-Kai Huang

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Basics

- 9+ years research experience in biomedical engineering which includes microscopy imaging, mammalian cell culture, drug treatment, assay development, and medical devices
- Solid hands-on experience with fluorescence, bioluminescence, and phosphorescence microscopy sample handling, external instrument installation, and image data analysis
- Demonstrated scientific excellence in the field by publishing 8 peer-reviewed articles and giving 3 oral presentations
- Independent, self-motivated, and fast learner with skill sets from multidisciplinary fields including electrical, mechanical, and biomedical engineering
- Strong mentor ability by training 20 microscope researchers and leading microscopy project update meetings

Education

University of California, Irvine

<https://www.uci.edu>

Doctor, Biomedical Engineering, Score: 3.96

Sep 2018–Jun 2023

Dissertation: Evaluating Mitochondria Transfer, Energy Metabolism, ROS Accumulation, and Development of Fluctuation Correlation using the side-SPIM

University of California, Irvine

<https://www.uci.edu>

Master, Biomedical Engineering

Sep 2018–Jun 2023

Thesis: Investigation of Epithelial Mitochondria Transfer via Fluorescence Lifetime Imaging Microscopy

National Taiwan Normal University

<https://www.ntnu.edu.tw>

Bachelor, Mechatronics Engineering, Score: 3.65

Sep 2012–Jun 2016

Work

Ellison Medical Institute

Microscopy Engineer

Jul 2023–Present

- Handled primary cells, cancer associated fibroblasts, and organoids in 3D culture including automatic seeding
- Developed label-free recognition neural network for live/dead colon and breast cancer organoid image analysis
- Developed user-friendly GUI application for batch FLIM data analysis using MATLAB and Python
- Maintain and document microscope (Operetta CLS, FV3000, VS200, and ECHO Revolve) SOPs in imaging center including instrument quality control, troubleshoot hardware dysfunction, and provide user training

University of California, Irvine

<https://www.uci.edu>

Graduate Student Researcher

Sep 2019–Jun 2023

- Handled breast cancer cell lines in 2D and built spheroids in 3D ECM such as matrigel and collagen
- Isolated pure mitochondria through chemical centrifugation and induced into breast cancer cell lines
- Operated microscopes for multiple applications such as fluorescent labelled mitochondria tracking in 3D using light sheet microscopy, protein interaction arc-EGFP using FLIM FRET, detect lipid droplets using third harmonic generation imaging, and determine diffusion coefficients of molecules using STICS
- Developed FLIM intensity-based image segmentation (FIBIS) algorithm to determine single mitochondria objects from low signal-to-ratio NADH signal and created analysis software for free and bound fraction of NADH lifetime

Skills

Microscopy: Expert **Keywords:** Fluorescence Microscopy, Laser Confocal Microscopy, FLIM, SPIM, TIRF, Particle Tracking, FCS, 2d-pcf, STICS, FRET, DIVER, STED, Slide Scanner, Micromanager

Experimental Lab: Expert

Keywords:

Cell Culture, Organoid/Spheroid Culture, Transfection, Plasmid Amplification, Mitochondria Isolation, MTT Assay, Seahorse Assay, Bradford Assay, Flow Cytometry, ImageJ, Spectrophotometer, Biosero Green Button Go, Eppendorf epMotion

Programming: Intermediate **Keywords:** MATLAB, Python, C/C#/C++, OpenCV, GUI Development, R, GraphPad Prism, Image Segmentation

Technical Drafting: Intermediate **Keywords:** Inventor, SolidWorks, MasterCAM, AutoCAD

Microprocessor & Hardware: Intermediate **Keywords:** Arduino, Raspberry Pi, 3D Printing

Publications

Communications Biology

Peristaltic forces drive tumor cell invasion in colorectal cancer Jan 2026
Curran A. Shah, Carly Strelez, Rachel Perez, **Yu-Kai Huang**, Christopher Cherry, Heinz-Josef Lenz, Shannon M. Mumenthaler. Under review.

Cell Systems

Systematic modeling of phenotypic drug response profiles in patient-derived organoids Jan 2026
Shannon Mumenthaler, Seungil Kim, Einar Bjarki Gunnarsson, Michael E. Doche, Yuyuan Zhou, **Yu-Kai Huang**, et al. Under review.

Journal of Cell Science

FLIM Intensity-Based Segmentation (FIBIS) Reveals Upregulated Energy Metabolism and Chemotherapy Sensitivity in MCF-7 Cells Jan 2025
Yu-Kai Huang, Mary Zhuang, Michelle A. Digman. Journal of Cell Science, 138 (23): jcs263702.

bioRxiv

<https://doi.org/10.1101/2024.05.24.595756>

Merging metabolic modeling and imaging for screening therapeutic targets in colorectal cancer May 2024
Niki Tavakoli, Emma J Fong, Abigail Coleman, **Yu-Kai Huang**, Mathias Bigger, Michael E Doche, Seungil Kim, Heinz-Josef Lenz, Nicholas Alexander Graham, Paul Macklin, Stacey D Finley, Shannon A Mumenthaler.

International Journal of Molecular Sciences

Breaking the concentration limit in fluorescence fluctuation spectroscopy with camera-based detection Aug 2022
Yu-Kai Huang, Per Niklas Hedde. Macromolecules. International Journal of Molecular Sciences 23 (17), 9840.

Environmental Science & Technology

<https://doi.org/10.1021/acs.est.3c01111>

Superoxide Release by Macrophages Dominates over Aqueous Chemistry of Secondary Organic Aerosols in Epithelial Lining Fluid
Ting Fang, **Yu-Kai Huang**, Jinlai Wei, Jessica Monterrosa Mena, Pascale Lakey, Michael Kleinman, Michelle Digman, Manabu Shiraiwa.

Frontiers in Synaptic Neuroscience

Palmitoylation-Regulated CaMKL Vesicle Interactions with Membranes and Arc/Arg3.1 Jan 2022
Barbara Barylko, Per Niklas Hedde, Clinton A. Taylor, Der D. Binns, **Yu-Kai Huang**, Gemma Molinaro, Kimberly M. Huber, David M. Jameson, and Joseph P. Albanesi. Volume 14.

IJCSIS

An Optimization Image Segmentation Clustering Algorithm for Badminton Sport Moving Tracking Applications from Video Sequences Jul 2020
Y. K. Huang, C. P. Huang. IJCSIS volume 15 No.7.

Projects

Mitochondria Transfer Between Cancer Cells and Normal Epithelial Cells

University of California, Irvine Sep 2019–Jun 2023

- Developed FLIM intensity-based image segmentation (FIBIS) algorithm for low signal to noise ratio intensity segmentation and created MATLAB graphic user interface available on GitHub (<https://github.com/KenHuang0820/FIBIS>)
- Quantified metabolic shift of single mitochondria compared with metabolic inhibitor drug treatment results using Agilent Seahorse XF ATP Real-Time rate assay achieving visualization and metabolism assay integration

Single Plane Illuminance Microscopy - Fluorescence Correlation Spectroscopy

University of California, Irvine Oct 2021–Mar 2022

- Fabricated home-made window chambers for side-SPIM microscope system and laser aligning of extracellular matrix substrate matching fraction index
- Designed image mean-square-displacement (iMSD) experiment applied in protein binding efficiency and live cell imaging to accomplish fluorescent correlation spectroscopy using light sheet

FLIM of Secondary Organic Aerosol (SOA) Induced to Macrophages

University of California, Irvine—in collaboration with Chemistry Dept. Oct 2020–Oct 2021

- Coordinated NAD(P)H phasor approach to FLIM data with superoxide production rate findings from collaborator and proved NADPH oxidase activation relate to SOAs and initiate lipid peroxidation
- Introduced Laurdan probe to quantify cell membrane fluidity disorder using lifetime difference for the first time and third harmonic generation imaging to visualize lipid peroxidation of SOAs

Volunteer

Biophysical Society Annual Meeting

Oral Presenter

Feb 2023–Present

"Metabolic Profiling of Transferred Mitochondria Using FLIM Intensity Based Image Segmentation (FIBIS)", Optical Microscopy and Super Resolution Imaging I, San Diego.

Biomedical Engineering Society (BMES) Annual Meeting

Oral Presenter

Oct 2022–Present

"Determining Metabolism Shift of Mitochondria Transfer using the Phasor Approach to FLIM", Biomedical Imaging and Instrumentation, San Antonio.

18th International Conference on Parallel and Distributed Computing, Applications and Technologies (PDCAT)

Oral Presenter

Dec 2017–Present

"A Depth-First Search Algorithm Based Otoloscope Application for Real-time Otitis Media Image Interpretation", Section Chair of Algorithms and Application, Taipei.